Galaga Repair Lab – Chat Summary

Lab 4: Vibe Coding · Repo: 31_galaga · AI used: Claude

Condensed summary of my two chats with the AI, not a word-for-word transcript. Session 1 times weren't saved.

Session 1: Bug fix and features

Student request

Shared the README and asked what the lab needs.

Assistant

Four tasks: fix the Bézier bug, then fill in `enemy_tint`, `on_wave_start` and `shield_charges`. Submit a 10 s before video, a 10 s after video and the chat history.

Student request

Uploaded the repo and asked for the bug fix and the three features.

Assistant

Task 1: the third term in `bezier()` was `3(1−t)t`, but it should be `3(1−t)t²`. The four weights should add up to 1. With the bug they add up to 1.375 at t = 0.5, so the enemies overshoot in the middle of the path. The start and end points still looked fine because the middle terms are zero at t = 0 and t = 1. Fixed by adding the missing `t` to the x and y lines.

Task 2: `enemy_tint` leaves wave 1 alone. After that it shifts each enemy colour's hue a bit more every wave. It only gets `kind` as input, so it uses a global `current_wave`.

Task 3: `on_wave_start` updates `current_wave` and shows "WAVE N" for 1.5 s. The timer goes down using `dt` in `update()`, so it doesn't depend on the frame rate.

Task 4: `shield_charges` returns `min(wave // 3, 3) or None`, which gives 1 charge every 3 waves with a max of 3. A ring around the ship and a "Shield N" label show the charges.

Student request

Asked how long the videos should be and what `self.wave` does.

Assistant

Both videos are 10 s. `self.wave` is the starting wave. Setting it to 3 for the recording shows the shield and new colours sooner, but it has to go back to 1 before submitting.

Student request

Sent the after video to check.

Assistant

The video shows the banner, the new colours and the shield. Trimmed it to 10 s.

Session 2: Checking the submission and committing (18 Oct 2026)

Student request 18:49

Shared the handout and both videos and asked if everything met the requirements.

Assistant

The before video was 16.6 s, so I cut it to 10 s (1 s to 11 s, skipping an old game-over screen). The after video was already 10 s. Still to do: export the chat, make a separate commit per task, upload to Lab 4 and set `self.wave` back to 1.

Student request 18:51

Asked how to make separate commits. `game.py` was still the original and `game_after.py` had all the changes.

Assistant

Copy one task at a time from `game_after.py` into `game.py`, then run `git add game.py` and `git commit`. Don't commit `.vscode/` or `game_after.py`.

Student request 18:54

Connected the project folder and asked the AI to make the edits one task at a time, without running git. The student would do the commits.

Assistant

`enemy_tint` needs `current_wave`, which only gets set in `on_wave_start`. So the Task 2 change also sets `current_wave`, and Task 3 adds the banner on top. That way the game runs after every commit. Made the Task 1 edit.

Student request 19:02

Asked where the commits would go.

Assistant

`origin` was the course repo (SETAPESU26/31_galaga). Committing only saves locally, so it's fine, but don't push there. Suggested putting the code in a personal repo and uploading the deliverables to the team repo's `Lab_04` folder.

Student request 19:05

Made a personal repo called `se_galaga`.

Assistant

Renamed the course remote to `upstream` and added `se_galaga` as `origin`. Checked with `git remote -v`.

Student request 19:12

Said the enemy colours hadn't changed after Task 2.

Assistant

That's expected on wave 1. The colours only change from wave 2. To test, start at wave 2 for a moment, then undo it with `git checkout -- game.py`.

Student request 19:12–19:19

Said `git commit` kept showing "nothing to commit" after the Task 2–4 edits.

Assistant

The file on disk still had the previous version. The AI's file-transfer step was sending an old copy. Using a new filename for each upload fixed it. Checked the final file was correct, and each task was committed.

Result

Commit	What changed
Fix Bezier curve: add missing t in third term	Fixed the third weight in `bezier()`
Add enemy_tint to recolour enemies each wave	Colours change each wave, using `current_wave`
Add wave banner in on_wave_start	"WAVE N" banner for 1.5 s
Add shield_charges: 1 charge every 3 waves	Shield charges, ring and label

Submitted: `galaga_before_10s.mp4`, `galaga_after_10s.mp4`, `game.py` (starts at wave 1) and this summary. Code: github.com/oomploompaluvr/se_galaga